Quantizable Models for M-theory

William D. Linch III

George P. and Cynthia W. Mitchell Institute for Fundamental Physics and Astronomy, Texas A&M University.

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(Based on work with Warren Siegel [1, 2, 3, 4].)

S-theory

- Pre-1984: There is a unique 11D supergravity and there are five 10D supergravities of types I, IIA/B, HE/O.
- Green & Schwarz (1984) [5]: 10D supergravities "$S(X)$" on spacetime X originate from quantization of fundamental strings "$\mathbf{S}(X)$" with target X.
- Moduli space of Type-II supergravity $S(X)$ on $X \to X \times T^n$:

$$\frac{O(n,n)}{O(n) \times O(n)}$$

  due to the symmetry between momentum modes (one copy of n) and winding modes (the other copy of n).
- Narain (1986) [6]: $O(n,n)$ symmetry of $S(X)$ broken to $O(n,n;\mathbf{Z})$ "T-duality" of the superstring $\mathbf{S}(X)$.

T-theory

- Duff (1989) [7]: String theories have a generalization that manifests T-duality at the cost of doubling the spacetime dimension $x \to (x, \tilde{x})$ for winding 0-modes.
- Siegel (1993) [8, 9, 10]: Doubled spacetime is reduced by a section condition on the momenta

$$(\partial_M) := \begin{pmatrix} \partial_\mu \\ \tilde{\partial}^\mu \end{pmatrix} \quad \eta := \begin{pmatrix} 0 & \delta^\nu_\mu \\ \delta^\mu_\nu & 0 \end{pmatrix} \quad \partial^M \partial_M = 0 \quad \Rightarrow \quad \text{e.g.} \quad \tilde{\partial}^m = 0.$$

Extension to a string theory $\mathbf{T}(X)$ with manifest T-duality symmetry. New Lie derivative (D-bracket) and commutator (C-bracket) described by current algebra of $O(D,D)$-covariant "generalized" momenta $\Pi_M = P_M + \eta_{MN} X'^N$ with quantum bracket

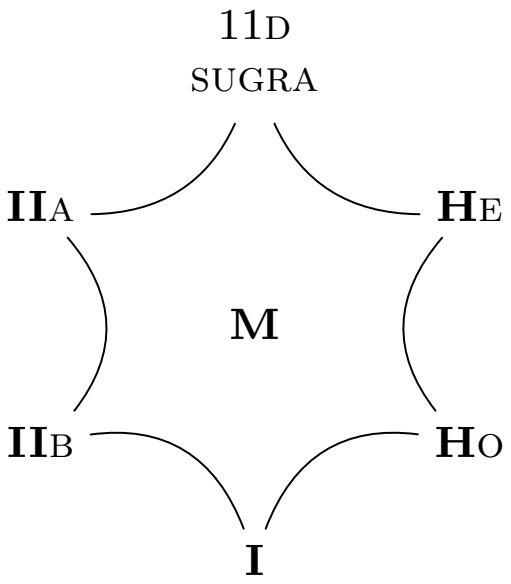
$$[\Pi_M(\sigma_1), \Pi_N(\sigma_2)] = 2i\eta_{MN}\delta'(\sigma_1 - \sigma_2).$$

Section condition arises dynamically as a new $O(D,D)$-invariant Virasoro constraint

$$L = \tfrac{1}{2}\eta^{MN}\Pi_M\Pi_N \quad \Rightarrow \quad \partial^M\partial_M = 0.$$

M-theory

- Cremmer, Julia, & Scherk (1978) [11]: Construction of eleven-dimensional supergravity. What about *its* fundamental origins?
- Bergshoeff, Sezgin, & Townsend (1987) [12, 13]: No strings in 11D but super-membranes. Strings do not explain existence of 11D supergravity but what are quantum membranes?
- Witten (1995) [14]: S-duality of the type-IIA **S**tring theory has 11-dimensional supergravity as its massless truncation.



11D Supergravity and U-duality

Insights into the fundamental structure of M-theory?

- Cremmer & Julia (1979) [15] and Hull & Townsend (1994) [16]: Analogously to the moduli space of compactifications of the type-II string theories on n-tori, hidden *exceptional* "U-duality" symmetry $E_{n(n)}$ of (gauged) 11D supergravity manifested by compactification on T^n.

n	$E_{n(n)}$	H_n	# vectors	# scalars
3	$SL(2;\mathbf{R}) \times SL(3;\mathbf{R})$	$SO(2) \times SO(3)$	6	7
4	$SL(5;\mathbf{R})$	$SO(5)$	10	14
5	$Spin(5,5)$	$SO(5) \times SO(5)$	16	25
6	$E_{6(6)}$	$Sp(4)$	27	42
7	$E_{7(7)}$	$SU(8)$	28	70
8	$E_{8(8)}$	$SO(16)$	0	128

- Duff (1985) [17]: "Either the $E_8 \times SO(16)$ structure appears only in $d = 3$ as an <u>internal</u> symmetry, or else it is really present at every $d \leq 11$ as a <u>spacetime</u> symmetry which transforms states of different spin into each other, in particular transforming [the frame field] into [the 3-form] in $d = 11$. We emphasize that we are advocating the latter."

F-theory?

Vafa (1996) [18]: $SL(2;\mathbf{Z})$ S-duality of the type-IIB **S**tring may be realized geometrically in "12D" theory F

$$\begin{tikzcd}
 & F(\widehat{X}) \arrow[dl, dotted] \arrow[dd] \\
M(X \times S^1) \arrow[dr] & \\
 & S(X)
\end{tikzcd}$$

Defining properties of F:

1. Geometrically realized (S(trong/weak) part of) U-duality
2. Not really 12D: Type-IIB **S**tring with axio-dilaton profile determined by elliptically-fibered Calabi-Yau 4-fold $T^2 \to \widehat{X} \to X$
3. Related by compactification to M-theory

Definition of $F(X)$

We propose to generalize the definition of F-theory by using both doubled and exceptional geometry [1]:
For any X, with supergravity $S(X)$, M-gravity $M(X)$, and generalized & doubled gravity $T(X)$, define $F(X)$ to make the following diagram commute:[1]

$$\begin{tikzcd}
 & \begin{array}{c} F(X) \\ E_{n(n)}/H_{n(n)} \end{array} \arrow[dl] \arrow[dr] & \\
\begin{array}{c} M(X) \\ \frac{GL(D+1,\mathbf{R})}{O(D,1)} \end{array} \arrow[dr, "\text{compactification}"'] & & \begin{array}{c} T(X) \\ \frac{O(D,D)}{O(D)\times O(D)} \end{array} \arrow[dl, "\text{sectioning}"] \\
 & \begin{array}{c} S(X) \\ \frac{GL(D,\mathbf{R})}{O(D-1,1)} \end{array} &
\end{tikzcd}$$

[1] The rank $n = D + 1$.

Extention to fundamental theories

Goal: For each rank $n = D + 1$ find brane theory (generalizing the string), the quantization of which gives the theory $\mathbf{F}(X)$ that completes the diagram

$$\begin{tikzcd}
 & \mathbf{F}(X) \arrow[dl, "\text{section}"'] \arrow[dr, "\text{reduction}"] & \\
\mathbf{M}(X) \arrow[dr] & & \mathbf{T}(X) \arrow[dl] \\
 & \mathbf{S}(X) &
\end{tikzcd}$$

It is *a priori* unclear whether this problem (existence of $\mathbf{F}$) has a solution.
In fact, we have the following

Result ([2, 3, 4])

Let $D = \dim(X) \leq 6$. Then $\mathbf{F}(X)$ exists in the form of a current algebra with exceptional global symmetry $G = E_{n(n)}$ of rank $n = D + 1$. This current algebra arises from the quantization of a free theory with Hamiltonian symmetry $H_{n(n)}$ the (split form of the) maximal compact subgroup of $E_{n(n)}$. When a Lagrangian formulation is known ($n \leq 5$), the symmetry is enhanced to $L_{n(n)}$ and the theory is that of a free self-dual form on a brane of split signature.

n	Lagrangian $L_{n(n)}$	Hamiltonian $H_{n(n)}$	Current Algebra $E_{n(n)}$
2	$SL(2;\mathbf{R})$	$SO(1,1)$	$GL(2;\mathbf{R})$
3	$SL(2;\mathbf{R})^2 \times SO(2)$	$SL(2;\mathbf{R}) \times SO(2)$	$SL(3;\mathbf{R}) \times SL(2;\mathbf{R})$
4	$SL(4;\mathbf{R}) \simeq Spin(3,3)$	$Sp(4;\mathbf{R}) \simeq Spin(3,2)$	$SL(5;\mathbf{R})$
5	$SL(4;\mathbf{C}) \simeq Spin(6;\mathbf{C})$	$Sp(4;\mathbf{C})$	$Spin(5,5)$
6	¿ $SU^*(8)$?	$USp(4,4)$	$E_{6(6)}$
7	¿ $U^*(8)$?	$SL(4;\mathbf{H})$	$E_{7(7)}$

Case Study: 3D String

Properties of the F-gravity diamond for case with X modeled on $\mathbf{R}^{2,1}$:

- Supergravity $S(X)$ is 3D, $N = 2$ supergravity; Lorentz group is $SO(2,1)$.
- Manifestly T-dual generalized & doubled SG has $SO(2,2) \sim SO(2,1)^2$.
- M-gravity $M(X)$ is 4D, $N = 1$ "old-minimal supergravity" (with real prepotential for conformal compensator) [19, 1].
- F-gravity $F(X)$ has exceptional U-duality symmetry of rank 4: $G = E_{4(4)} \simeq SL(5;\mathbf{R})$.
- Coordinates X^{mn} in the **10** (signature 4+6). But solution of sectioning

$$\varepsilon^{mnpqr}\partial_{mn}\partial_{pq} = 0 \quad \Rightarrow \quad (\partial_{mn}) = (\partial_{-1\mu}, \partial_{\mu\nu}) \to (\partial_\mu, 0)$$

  reduces this to $(3+1)$-dimensional $M(X)$ breaking $SL(5;\mathbf{R}) \to GL(4;\mathbf{R})$ spontaneously [20].

Virasoro currents

Strategy: Attempt to interpret strong constraint as a Virasoro-type constraint for the Hamiltonian analysis of some worldvolume theory.

- In $\mathbf{T}(X)$ we saw the left-moving worldsheet current

$$\Pi_M(\sigma) = P_M + \eta_{MN} X'^N \quad \text{with} \quad [P_M(\sigma_1), X^N(\sigma_2)] = -i\delta_M^N \delta(\sigma_1 - \sigma_2)$$

  and Virasoro constraint $L = \frac{1}{2}\eta^{MN}\Pi_M\Pi_N$ giving sectioning $\partial^M\partial_M = 0$.
- So we postulate for $\mathbf{F}(X)$ at each fixed τ a wordvolume with 5 coordinates σ^m (in addition to the evolution coordinate τ) and $G = SL(5;\mathbf{R})$-covariant current

$$\Pi_{mn} := P_{mn} + \tfrac{1}{2}\varepsilon_{mnpqr}\partial^r X^{pq}.$$

- Virasoro-type generator for strong constraint on string fields:

$$\mathcal{S}^r := \tfrac{1}{16}\varepsilon^{mnpqr}\Pi_{mn}\Pi_{pq}.$$

- Implies $\varepsilon^{mnpqr}(\partial_{mn} f)(\partial_{pq} g) = 0$ for all functions f and g on $F(X)$.

Problem ... and solution

But now algebra does not close:

$$[\mathcal{S}^m, \mathcal{S}^n] = \mathcal{S}^{(m}\,\partial^{n)}\delta + \partial^{[m}\mathcal{S}^{n]}\,\delta + \varepsilon^{mnpqr}\Pi_{pq}\mathcal{U}_r\,\delta \quad \text{with} \quad \mathcal{U}_r := \partial^s\Pi_{rs}.$$

$\Rightarrow$ Must interpret $\mathcal{U}$ as new constraint. It is central and reduces to

$$\mathcal{U}_m = \partial^n P_{mn}$$

... but this is a Gauß law constraint! Analogously to the first-quantization of Maxwell theory [21]: Quantize a 6D gauge 2-form X_{MN}!

	gauge field	Lagrange multiplier	dynamical components	conjugate momentum	Gauß's law
Maxwell	A_μ	A^τ	A^i	E^i	$\partial^i E_i$
$\mathbf{F}(X)$	X_{MN}	$X^{\tau m}$	X^{mn}	P_{mn}	$\partial^n P_{mn}$

Chiral forms

Siegel (1983) [22]: By analogy with string action for chiral scalar fields, action for gauge 2-form with self-dual fieldstrength F_{MNP}:

$$S_{iegel} = \int \left(\tfrac{1}{12} F^{MNP} F_{MNP} + \tfrac{1}{4} \lambda^{MN} F_M^{(-)PQ} F^{(-)}_{NPQ} \right) d^6\sigma.$$

The equation of motion for λ

$$F_M^{(-)PQ} F^{(-)}_{NPQ} = 0 \quad \Rightarrow \quad F^{(-)}_{MNP} = 0$$

implies the fieldstrength of X is self-dual. The gauge transformations

$$\delta X_{MN} = \alpha^P F^{(-)}_{MNP} + \tfrac{1}{12} \alpha^P \lambda_{[M}{}^Q F(-)_{NPQ]} + \mathcal{O}(\lambda^2)$$
$$\delta \lambda^{MN} = \partial^{(M} \alpha^{N)} + \alpha^P \partial_P \lambda^{MN} + \tfrac{1}{2} \lambda_P{}^{(M} \left(\partial^{N)} \alpha^P - \partial^P \alpha^{N)} \right) - \text{trace} + \mathcal{O}(\lambda^2)$$

imply λ is pure gauge.

Canonical analysis

In $\lambda \to 0$ gauge the Hamiltonian action becomes

$$S = -\int \tfrac{1}{2}\dot{X}^{mn} P_{mn}\, d^5\sigma d\tau + \int H\, d\tau$$

$$H = \int \left(\tfrac{1}{4}P^{mn}P_{mn} + \tfrac{1}{12}F^{mnp}F_{mnp} + X^{\tau m}\mathcal{U}_m\right)\, d^5\sigma.$$

Current from selfdual fieldstrength:

$$\Pi_{mn} := F^{(+)}_{\tau mn} = P_{mn} + \tfrac{1}{2}\varepsilon_{mnpqr}\partial^r X^{pq}.$$

Free quantum brackets $[P_{mn}(\sigma_1), X^{pq}(\sigma_2)] = -i\delta^{[p}_{[m}\delta^{q]}_{n]}\delta(\sigma_1 - \sigma_2)$ imply

$$\boxed{[\Pi_{mn}(\sigma_1), \Pi_{pq}(\sigma_2)] = 2i\epsilon_{mnpqr}\partial^r\delta(\sigma_1 - \sigma_2)}$$

Stress-energy tensor

$$\mathcal{T}_{mn} = \eta_{mn}\mathcal{T} - \tfrac{1}{2}\eta^{pq}\Pi_{mp}\Pi_{nq} \qquad \boxed{\mathcal{S}^r = \tfrac{1}{16}\epsilon^{mnpqr}\Pi_{mn}\Pi_{pq}} \qquad \mathcal{T} = \tfrac{1}{8}\eta^{mp}\eta^{nq}\Pi_{mn}\Pi_{pq}$$

$\Rightarrow$ Closed $E_{4(4)} = SL(5;\mathbf{R})$-covariant sub-algebra generated by $\mathcal{S}$ and $\mathcal{U}$!

Courant bracket

Spacetime gauge transformations generated by vector field currents

$$\Lambda_i := \tfrac{i}{2} \int \lambda_i^{mn}(X(\sigma))\Pi_{mn}(\sigma)\, d^5\sigma.$$

Algebra

$$[\Lambda_1, \Lambda_2] = \Lambda_{12} \quad \text{with} \quad \lambda_{12}^{mn} = \tfrac{1}{4}\lambda_{[1}^{pq}\partial_{pq}\lambda_{2]}^{mn} - \tfrac{1}{2}\lambda_{[1}^{p[m}\partial_{pq}\lambda_{2]}^{n]q} - \tfrac{1}{4}\lambda_{[1}^{mn}\partial_{pq}\lambda_{2]}^{pq}$$

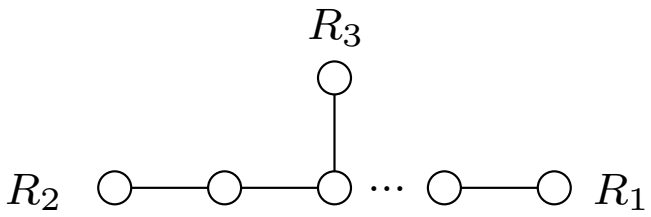
exactly reproduces the "exceptional Courant bracket" of diffeomorphisms of the $E_{4(4)}$ F-gravity computed by Berman, Godazgar,[2] and Perry [23].

Conclusion:

1. Quantization of the chiral 2-form on a 5-brane with split signature gives rise to a current algebra with $E_{4(4)}$ symmetry.
2. The massless truncation of its C-bracket reproduces the "exceptional Courant bracket" of diffeomorphisms of the M-gravity appropriate to this case.
3. Solution of the constraints reduces the current algebra to that of the three-dimensional bosonic string.

U-duality

The U-duality $E_{4(4)}$ can be enlarged to $G = E_{n(n)}$.
Consider the Dynkin diagram



(This corresponds to $R_1 = \mathbf{10}'$, $R_2 = \mathbf{5}$, and $R_3 = \mathbf{5}'$ for $E_{4(4)} = SL(5;\mathbf{R})$.)
Let $\mathbf{m}$ run over R_1) and m run over R_2 and introduce the $(R_1 \otimes R_1)_{sym} \supset R_2$ Clebsch-Gordan coefficients $\eta_{\mathbf{mn}p}$ and $\eta^{\mathbf{mn}p}$.
Generalization of the current algebra:

$$[\Pi_{\mathbf{m}}(\sigma_1), \Pi_{\mathbf{n}}(\sigma_2)] = 2i\eta_{\mathbf{mn}p}\partial^p\delta(\sigma_1, \sigma_2)$$

Virasoro constraint:

$$\mathcal{S}^p = \tfrac{1}{4}\eta^{\mathbf{mn}p}\Pi_{\mathbf{m}}\Pi_{\mathbf{n}}.$$

C-bracket of two vector fields:

$$[V_1^{\mathbf{m}}\Pi_{\mathbf{m}}(\sigma_1), V_2^{\mathbf{n}}\Pi_{\mathbf{n}}(\sigma_1)] = 2i\eta_{\mathbf{mn}p}\partial^p\delta(\sigma_1-\sigma_2)V_1^{\mathbf{m}}V_2^{\mathbf{n}} - i\delta(\sigma_1-\sigma_2)\left[V_{[1}^{\mathbf{m}}\partial_{\mathbf{m}}V_{2]}^{\mathbf{n}} - \tfrac{1}{2}\eta^{\mathbf{mn}r}\eta_{\mathbf{pq}r}V_1^{\mathbf{p}}\partial_{\mathbf{m}}V_2^{\mathbf{q}}\right]\Pi_{\mathbf{n}}.$$

Truncation to 0-modes should reproduce the $E_{n(n)}$ "Courant brackets" computed by Berman, Cederwall, Kleinschmidt, and Thompson [20].
Therefore, the projector $(R_1 \otimes R_1)_{sym} \to R_2$ is identified with the Clebch-Gordan coefficients:

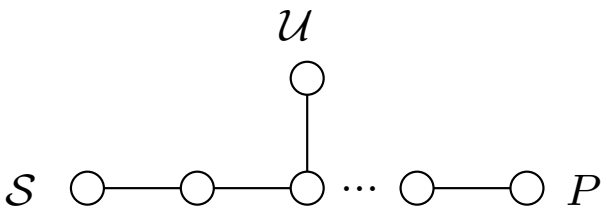
$$Y^{\mathbf{mn}}{}_{\mathbf{pq}} \leftrightarrow \eta^{\mathbf{mn}r}\eta_{\mathbf{pq}r}$$

up to constraints.
This Y projector factorizes for all ranks $n \leq 6$ and factorizes up to a new constraint for $E_{7(7)}$ (corresponding to the 6D string).
Ultimate goal is 10-dimensional Lorentz invariance for X corresponding to E_{11} U-duality conjectured by West (2001) [24].

Open questions and current & future work

Loads of work to do...

1. Relation to M2 and M5 in 11D target space of Bergshoeff, Sezgin, & Townsend (1987) [12, 13] and Howe & Sezgin (1996) [25, 26]?
2. More U-duality? (Can we use Palmkvist [27] and related work to get $E_{>7}$?)
3. Narian-type description in terms of exceptional Borcherds algebras?



4. Relation to (gravitational) tensor hierarchies?
5. Worldvolume gravity?
6. Quantization?

THANK YOU!

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